

Project Phase 1: Problem Definition and Dataset Creation

A Supervised Machine Learning Approach for Satellite-Based Surface Monitoring in Hunga
Tonga–Hunga Ha’apai

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Abstract

This first phase defines the case study, study area, satellite image, and dataset construction strategy for a supervised land-cover classification project based on Sentinel-2 imagery. The selected region of interest corresponds to the Hunga Tonga–Hunga Ha’apai volcanic system in the South Pacific Ocean. Although the broader case study is associated with the January 2022 volcanic eruption, this phase uses a pre-eruption Sentinel-2 image acquired on November 18, 2021, because it preserves a visible landmass and contains enough spectral variability to support supervised classification. The proposed dataset is built from manually digitized training polygons representing five surface classes: ocean water, bare soil, vegetation, cloud, and cloud shadow. The pixel values extracted from ten Sentinel-2 spectral bands will be transformed into a tabular TSV file containing geographic coordinates, spectral signatures, and class labels. This dataset will be used in subsequent phases to train and compare multiple machine learning architectures.

1 Introduction

Modern geospatial analysis increasingly relies on satellite imagery and machine learning methods to classify, monitor, and quantify land-cover patterns. In this project, the objective is to construct a supervised classification workflow using multispectral Sentinel-2 imagery. The workflow transforms raster-based spectral information into a tabular machine learning dataset and prepares the input data required for model training, evaluation, and spatial prediction.

The selected case study is the Hunga Tonga–Hunga Ha’apai volcanic system. This region is especially relevant because of the major eruption that occurred in January 2022, which caused severe geomorphological changes, including land loss, coastline modification, and volcanic material redistribution. However, for this first classification stage, the project uses a pre-eruption image from November 18, 2021. This image was selected because the island is still clearly visible and because the scene includes several distinguishable surface conditions, such as ocean water, exposed land, vegetation, clouds, and cloud shadows.

This document corresponds to Phase 1 of the final project and focuses on problem definition, study area description, satellite imagery metadata, class definition, and dataset creation strategy.

2 Problem Definition

The general problem addressed in this project is the classification of surface cover types in a multispectral satellite image using supervised machine learning. The selected image contains both terrestrial and atmospheric features, which makes it suitable for evaluating whether different machine learning models can distinguish spectral patterns associated with land, vegetation, water, clouds, and cloud shadows.

The classification problem can be formulated as a pixel-level supervised learning task. Each pixel is represented by a spectral signature composed of Sentinel-2 band values. The target variable corresponds to the manually assigned land-cover class derived from training polygons.

3 Study Area

The region of interest corresponds to the Hunga Tonga–Hunga Ha’apai volcanic system, located in the South Pacific Ocean, north of Tongatapu, the main island of the Kingdom of Tonga. The area includes the volcanic island and its surrounding marine environment.

This site was selected because it represents an important geophysical case study and because previous analysis of the same region showed strong spatial changes associated with the January 2022 eruption. For the present classification task, the image acquired before the eruption was selected to ensure a greater amount of visible land surface, clouds, and more balanced class representation.

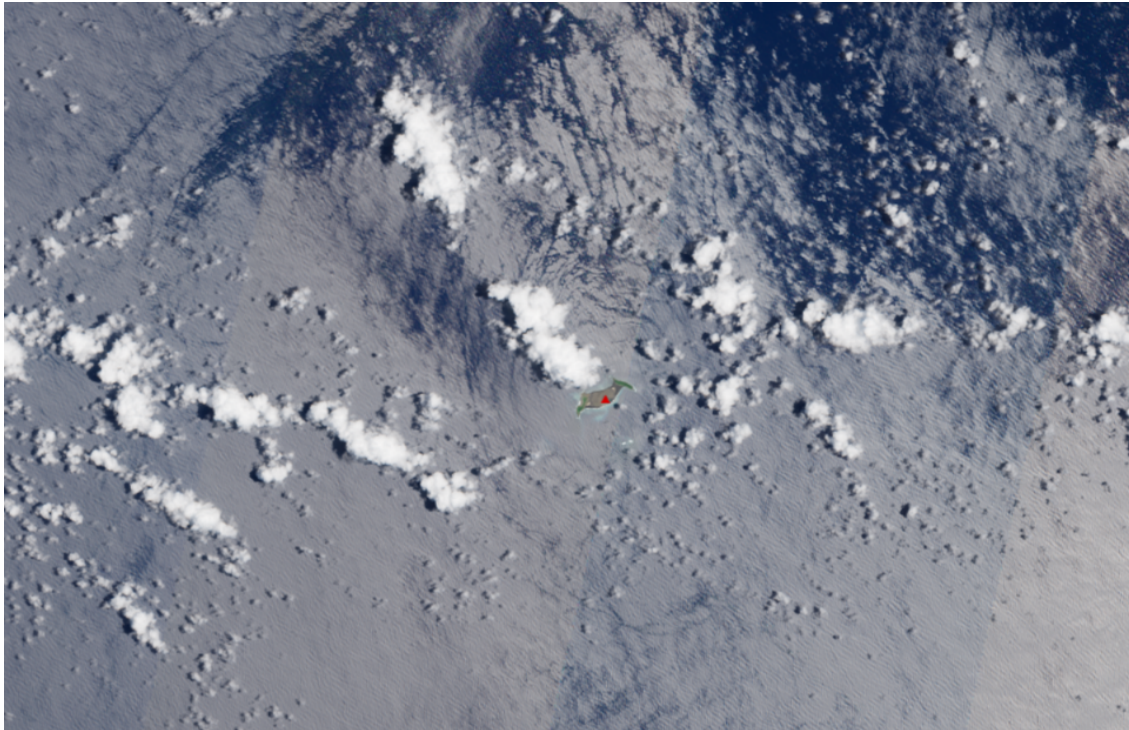


Figure 1: Geographic location of the Hunga Tonga–Hunga Ha’apai study area.

4 Selected Event

The broader environmental context of the project is the Hunga Tonga–Hunga Ha’apai volcanic eruption of January 2022. This eruption dramatically modified the volcanic island and surrounding coastal environment. The event is relevant for remote sensing because satellite imagery allows the observation and quantification of land-cover and coastline changes before and after the eruption.

However, the current phase does not perform change detection. Instead, it uses the case study as a basis for a supervised classification exercise. The selected pre-eruption image provides a more complete landmass and allows the classification of multiple land-cover and atmospheric classes.

5 Satellite Image Metadata

The selected image corresponds to Sentinel-2 multispectral imagery acquired before the major eruption. Sentinel-2 imagery was selected because it provides high spatial resolution optical bands and multiple spectral channels suitable for land-cover classification.

Table 1: Satellite image metadata.

Metadata item	Description
Satellite mission	Sentinel-2
Product type	Level-2A surface reflectance product
Acquisition date	November 18, 2021
Region of interest	Hunga Tonga–Hunga Ha’apai
Coordinate reference system	EPSG:32701 – WGS 84 / UTM zone 1S
Primary spatial resolution	10 m
Image use	Supervised land-cover classification
Main processing environment	Python and QGIS

6 Spectral Bands

The dataset uses ten Sentinel-2 spectral bands. Four bands are originally available at 10 m resolution, while six bands are available at 20 m resolution and will be resampled to the 10 m reference grid during the Python processing workflow.

Table 2: Sentinel-2 bands selected for dataset construction.

Band	Description	Original resolution
B02	Blue	10 m
B03	Green	10 m
B04	Red	10 m
B05	Red Edge 1	20 m
B06	Red Edge 2	20 m
B07	Red Edge 3	20 m
B08	Near Infrared	10 m
B8A	Narrow Near Infrared	20 m
B11	Short-Wave Infrared 1	20 m
B12	Short-Wave Infrared 2	20 m

The 10 m bands are used as the spatial reference. The 20 m bands are resampled to match the 10 m grid, ensuring that all spectral values correspond to the same pixel locations.

7 Visualization Composites

Two visual composites were prepared in QGIS to support the interpretation and digitization of training polygons.

7.1 Natural Color Composite

The natural color composite was created using bands B04, B03, and B02 assigned to the red, green, and blue channels respectively. This visualization approximates a true-color image and supports the identification of ocean water, exposed land, clouds, and cloud shadows.

Table 3: Natural color composite.

Display channel	Sentinel-2 band
Red	B04
Green	B03
Blue	B02

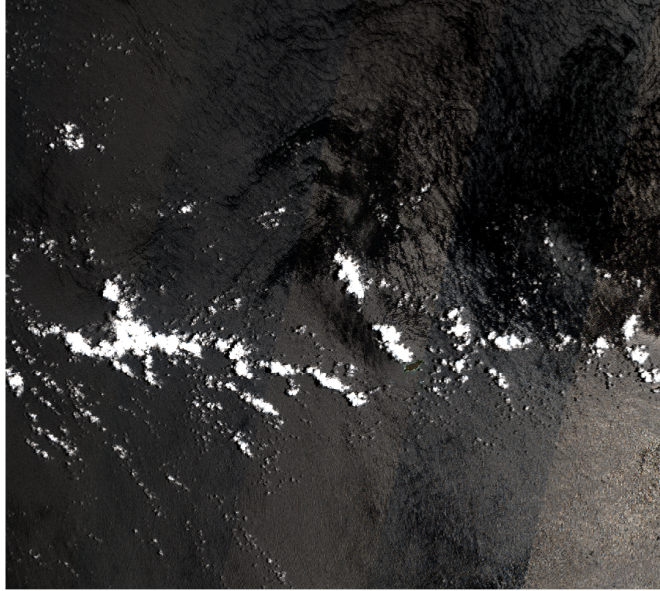


Figure 2: Natural color composite of the selected Sentinel-2 image.

7.2 False Color Composite

The false color composite was created using bands B08, B04, and B03 assigned to the red, green, and blue channels respectively. This visualization is useful for detecting vegetation because healthy vegetation strongly reflects near-infrared radiation and appears in red tones.

Table 4: False color composite.

Display channel	Sentinel-2 band
Red	B08
Green	B04
Blue	B03

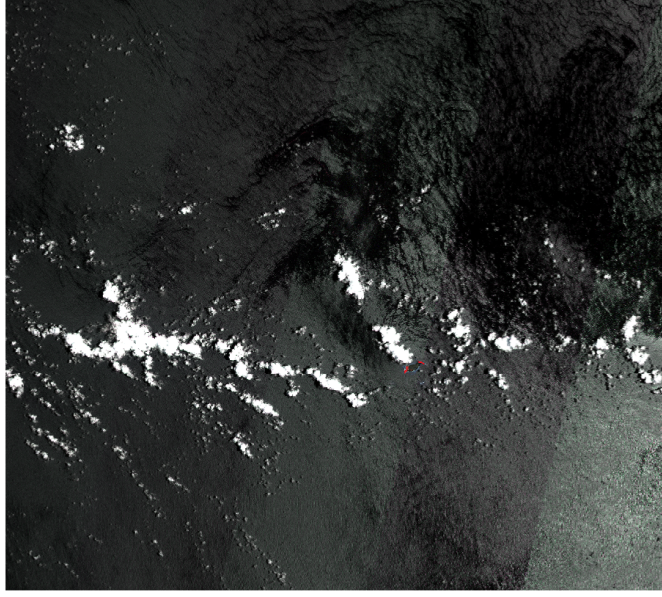


Figure 3: False color composite used to support vegetation identification.

8 Class Definition

Five classes were defined for the supervised classification task. These classes represent the main surface and atmospheric conditions visible in the selected image.

Table 5: Land-cover and atmospheric classes used in the training dataset.

Class ID	Class name	Description
1	Ocean_water	Open ocean water surrounding the island.
2	Bare_soil	Exposed land, volcanic soil, sand, or non-vegetated terrestrial surface.
3	Vegetation	Vegetated areas identified mainly through false color interpretation.
4	Cloud	Bright cloud features visible in the optical image.
5	Cloud_shadow	Dark shadow areas associated with cloud cover.

The inclusion of cloud and cloud shadow classes is important because the selected image contains atmospheric features. If these features were not labeled, the machine learning models could confuse them with bright land surfaces or dark water areas.

9 Training Polygon Creation

Training polygons were manually digitized in QGIS. Each polygon represents a visually interpreted sample of one class. The polygons were created over homogeneous areas to reduce mixed-pixel effects, especially near coastlines, cloud borders, and transitions between vegetation and bare soil.

The polygon layer was stored as a GeoJSON file named:

`training_polygons2.geojson`

The attribute table includes at least the following fields:

- **class_id**: numerical identifier of the class.
- **class_name**: categorical class label.

Table 6: Number of training polygons by class.

Class name	Number of polygons
Ocean_water	17
Cloud	14
Bare_soil	14
Vegetation	14
Cloud_shadow	14

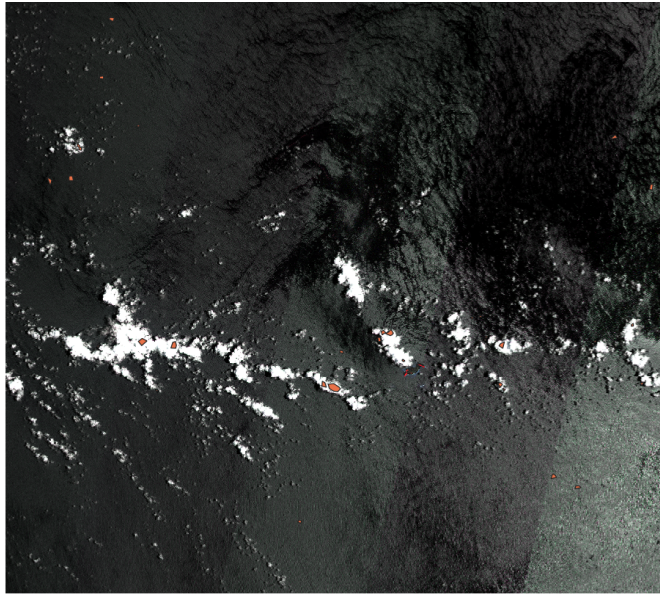


Figure 4: Training polygons digitized over the selected Sentinel-2 image.

10 Dataset Creation Strategy

The dataset creation process consists of transforming the raster-based Sentinel-2 bands into a non-spatial tabular file suitable for machine learning. This is done by extracting the spectral values of the pixels located inside each training polygon.

The extraction process is implemented in Python using geospatial and data processing libraries. The main steps are:

1. Load the Sentinel-2 spectral bands.
2. Use B02 at 10 m resolution as the reference grid.
3. Resample the 20 m bands to match the 10 m spatial grid.
4. Load the training polygons from the GeoJSON file.
5. Extract all pixel values located within the polygons.

6. Convert projected coordinates to latitude and longitude.
7. Assign each pixel the corresponding `class_id` and `class_name`.
8. Export the final table as a TSV file.

The expected output file is:

`training_data.tsv`

11 TSV Structure

The resulting TSV file contains one row per extracted pixel. Each row stores the geographic location of the pixel center, the spectral values from the selected Sentinel-2 bands, and the class label inherited from the training polygon.

Table 7: Expected structure of the training dataset.

Column	Description
Latitude	Pixel center latitude in geographic coordinates.
Longitude	Pixel center longitude in geographic coordinates.
B02	Blue band reflectance value.
B03	Green band reflectance value.
B04	Red band reflectance value.
B05	Red Edge 1 band reflectance value.
B06	Red Edge 2 band reflectance value.
B07	Red Edge 3 band reflectance value.
B08	Near Infrared band reflectance value.
B8A	Narrow Near Infrared band reflectance value.
B11	Short-Wave Infrared 1 band reflectance value.
B12	Short-Wave Infrared 2 band reflectance value.
class_id	Numerical class identifier.
class_name	Categorical class label.

12 Dataset Balance Considerations

Since ocean water and clouds can occupy large areas in the image, the number of extracted pixels may vary significantly across classes. To prevent model bias in later phases, the dataset will be balanced before model training by sampling an equal number of pixels per class. This will allow the comparison of machine learning architectures under more consistent conditions.

Conclusion

This first phase establishes the foundation for the supervised classification workflow. The Hunga Tonga–Hunga Ha’apai volcanic system was selected as the case study, and a Sentinel-2 pre-eruption image acquired on November 18, 2021 was chosen because it provides sufficient visual and spectral variability for land-cover classification. Five classes were defined: ocean water, bare soil, vegetation, cloud, and cloud shadow. Training polygons were created in QGIS and will be used to extract pixel-level spectral signatures from ten Sentinel-2 bands. The resulting TSV dataset will support the next project phase, which focuses on preprocessing, machine learning model training, and initial performance evaluation.